

3) Consider three different raster systems with resolutions of 640 by 480, 1280 by 1024, and 2560 by 2048. Assuming 8 bits per byte, what size frame buffer (in bytes) is needed for each of these systems to store 12 bits per pixel? Assuming 8 bits per byte, what size frame buffer (in bytes) is needed for each of these systems to store 24 bits per pixel?

To obtain the size of the frame buffer needed we first multiply the resolution values of the raster system with the amount of pixels to be stored, and then divide it by the 8 bits per byte. This is done as follow:

Frame buffer is needed for each of these systems to store 12 and 24 bits per pixel:

a) to store 12 bits per pixel:

$$640 \times 480 \times 12 \text{ bits} \div 8 \text{ bits per byte} = 450 \text{ KB}$$

$$1280 \times 1024 \times 12 \text{ bits} \div 8 \text{ bits per byte} = 1920 \text{ KB}$$

$$2560 \times 2048 \times 12 \text{ bits} \div 8 \text{ bits per byte} = 7680 \text{ KB}$$

b) to store 24 bits per pixel:

$$640 \times 480 \times 24 \text{ bits} \div 8 \text{ bits per byte} = 2 \times 450 \text{ KB}$$

$$1280 \times 1024 \times 24 \text{ bits} \div 8 \text{ bits per byte} = 2 \times 1920 \text{ KB}$$

$$2560 \times 2048 \times 24 \text{ bits} \div 8 \text{ bits per byte} = 2 \times 7680 \text{ KB}$$

4) How long would it take to load a 640 by 480 frame buffer with 12 bits per pixel, if 10^5 bits can be transferred per second? How long would it take to load a 24 bit per pixel frame buffer with a resolution of 1280 by 1024 using this same transfer rate?

To calculate the time needed to load certain frame size buffer with 12 bits per pixel we first find out its frame size to then divide by the amount of bits being transferred as shown below.

a) 12 bits per pixel frame buffer:

$$\text{Total number of bits for the frame} = 640 \times 480 \times 12 \text{ bits} = 3,686,400 \text{ bits}$$

$$\text{The time needed to load the frame buffer} = 3686400 / 10^5 \text{ sec} = 36.864 \text{ sec}$$

b) 24 bits per pixel frame buffer.

$$\text{Total number of bits for the frame} = 1280 \times 1024 \times 24 \text{ bits} = 31,457,280 \text{ bits}$$

$$\text{The time needed to load the frame buffer} = 31457280 / 10^5 \text{ sec} = 314.5728 \text{ sec}$$

Computer Graphics I
Homework #2: Frame Buffers and Display Systems

Date due: March 21st 2013
Author: Piter Garcia

5) How much time is spent scanning across each row of pixels during screen refresh on a raster system with a resolution of 1280 by 1024 and a refresh rate of 60 frames per second.

To obtain the time spent in scanning each row of pixel we simply divide one by the rate of each pixel, now to figure out this rate we multiply the frame time by its minimum resolution value. This is done as follows:

a-1) The scan rate for each pixel row is;
 $60 \text{ frames/sec} \times 1024 \text{ lines/frame} = 61,440 \text{ lines/sec}$

a -2) Scan time per frame is;
 $1/60 \text{ sec}$, or approximately 16.7 milliseconds

The scan time is approximately:
16.3 microseconds per scan line.